

AJINKYA GOKHALE

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Education

University of Stuttgart

MSc. Infotech, Major: Computer Hardware/Software Engineering

Oct. 2023 – Present

Stuttgart, Germany

Veermata Jijabai Technological Institute, Mumbai

Bachelor of Technology in Electronics and Telecommunications Engineering (GPA: 8.32/10)

July 2020 – May 2023

Mumbai, India

Government Polytechnic Amravati

Diploma in Electronics and Telecommunications Engineering (96.68%)

July 2017 – May 2020

Amravati, India

Experience

Nineti GmbH (Startup)

Working Student (Former Intern)

Feb. 2024 – Present

Stuttgart, Germany

- **Founding Engineer:** Spearheaded development of Stromleser IoT devices, mobile application, and **AWS backend infrastructure**, contributing to the company's foundational technology stack.
- **AWS Cloud Solutions:** Implemented scalable cloud infrastructure managing data from **10,000+ IoT devices** using EC2, Lambda, DynamoDB, and IoT Core, ensuring 99.9% uptime and real-time data processing.
- **Hardware Design:** Designed and tested **20+ electronic circuits** using KiCad, taking them from schematic to production-ready PCBs for battery-operated IoT devices.
- **Firmware Development:** Developed embedded firmware based on **ESP-IDF**, implementing the network stack with IP communication, **MQTT-based messaging to AWS IoT Core**, and Home Assistant integration.

Nomadists India Pvt. Ltd.

Full Stack Developer Intern

May 2020 – Aug. 2020

Pune, India

- Created REST APIs with **Node.js and Express.js** and designed the **PostgreSQL** database schema using Sequelize ORM.
- Wrote server-side logic for user authentication and data validation and integrated third-party APIs into the backend.
- Built the frontend in **React.js** with responsive layouts and connected it to the backend through REST endpoints.
- Used **Git and GitHub** for version control and followed agile sprints with code reviews and CI/CD through GitHub Actions.

Projects

SwapMails - AWS Serverless Email Platform | Full-Stack Web App and Chrome Extension Jan. 2025 – Present

- Built the entire platform on **AWS Lambda and SES and DynamoDB** using CDK v2 for infrastructure as code. It serves **1000+ users** with 99.9% uptime.
- Set up email processing that is **GDPR-compliant** with automatic TTL cleanup in DynamoDB and Lambda-based parsing. Emails arrive in under 60 seconds.
- Published a **Chrome extension** on the Chrome Web Store that has **500+ active users**. It generates emails in one click and syncs the inbox in real time.
- Built the frontend in **React.js** with real-time updates and multi-language support and deployed it globally through AWS CloudFront.
- Created the REST API using **AWS API Gateway** with authentication and rate limiting. It handles **10,000+ API calls** per day.
- Added AES-256 encryption and automatic data deletion and audit logging to make the platform privacy-first by design.
- Cut hosting costs by **95%** compared to running on traditional servers by going fully serverless. The system handles traffic spikes up to 1000x the normal load.

Gesture Recognition using Time-of-Flight Camera | Python, OpenCV, Computer Vision Jan. 2023

- Built a hand gesture recognition system in **Python with OpenCV** using a Time-of-Flight depth camera. It detects gestures in real time at **95% accuracy**.
- Wrote custom filtering and classification logic that reduced false positives by **20%** compared to the baseline approach.
- Integrated the gesture system as a control interface so users could interact with applications using hand movements instead of a mouse.

Technical Skills

Cloud Platforms: AWS (Expert) - Lambda, EC2, S3, DynamoDB, RDS, CloudFormation, CloudWatch, IAM, VPC, API Gateway, SES, IoT Core — Google Cloud Platform

Programming Languages: Python, Java, JavaScript, C/C++, Rust, HTML5/CSS3, TypeScript

Backend & APIs: Node.js, Express.js, RESTful APIs, Microservices Architecture, Serverless Computing, Event-Driven Architecture

Frontend Development: React.js, HTML5, CSS3, JavaScript, Chrome Extension Development, Responsive Design

Infrastructure as Code: AWS CDK v2, CloudFormation, Terraform — CI/CD Pipelines - GitHub Actions, AWS CodePipeline, Jenkins

Databases: Amazon DynamoDB, Amazon RDS (PostgreSQL, MySQL), SQL, NoSQL, Database Design, Data Modeling

DevOps & Monitoring: Docker, AWS CloudWatch, X-Ray, Linux Administration, Performance Monitoring, Log Analysis

IoT & Embedded Systems: AWS IoT Core, MQTT, Raspberry Pi, Arduino, ESP32/NodeMCU, Sensor Integration, Firmware Development

Development Tools: VS Code, Git, GitHub, AWS CLI, Postman, Jira, KiCad

Security & Compliance: GDPR Compliance, AWS IAM, SSL/TLS, Network Security, Data Privacy, Encryption

Languages

English: Professional Proficiency

German: Intermediate (B1 Level)

Hindi: Native Speaker

Hobbies

Local LLMs: Optimizing inference, testing quantized models, and exploring open-source AI on personal hardware.

Self-Hosting: Running services like Home Assistant and Nextcloud on a personal Kubernetes cluster.

Homelab: Tinkering with container orchestration, networking, and infrastructure automation.

Extracurricular Activities

GPHReviews - Tech Content Creator

Digital Content Creator & Community Builder

- Grew a YouTube channel to **42,000+ subscribers** by publishing regular tech reviews and tutorials over 6 years.
- Partnered with **iOS app developers** and tech companies to review products and run promotional campaigns.
- Built an online community of Apple tech enthusiasts and hosted discussions around new product launches.

Spring 2017 – Oct. 2023

YouTube Platform

Certifications

- AWS Certified Solutions Architect
- AWS Certified AI Practitioner